PH4102: Introductory Astrophysics

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Nucleosynthesis inside star

Primary component of newly born stars is hydrogen. Other heavier elements may also be present initially, but fractions is assumed to be very less.

Through nuclear *fusion* reactions, heavier elements from the initial hydrogen is formed.

We discuss below a few well understood chain of reactions which, we believe, are operating inside stars and are responsible for producing heavier elements from hydrogen.

Hydrogen to Helium:

The two set of chain reactions for this: Proton-proton chain (pp chain) and Carbon-Nitrogen-Oxygen (CNO) cycle.

Helium to Carbon:

Triple alpha process.

Further reactions:

Carbon $\longrightarrow$ Oxygen $\longrightarrow$ Silicon $\longrightarrow$ Iron

Intermediate elements are also formed during the process of forming above materials.

Hydrogen to Helium:

pp chain :

Proposed by Bethe and Critchfield (1938).

$$^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{2}_{1}\mathrm{H} + e^{+} + \nu_e$$

$$^{2}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + \gamma$$

Intermediate deuterium and helium-3.

The first reaction step has a very small probability (because of weak interaction involvement). At the central density and temperature of the Sun, the expected time for a proton to collide with another one to form a deuteron is 10^{10} years on the average. It is only thanks to the slowness of this reaction that the Sun is still shining.

After the production of helium-3, the reactions can proceed through three alternative branches: *pp*1, *pp*2, *pp*3. The branch *pp*1 is by far the dominant branch (69%) for conditions corresponding to the Solar interior.

pp1:

$$^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} \rightarrow {}^{4}_{2}\mathrm{He} + 2\ {}^{1}_{1}\mathrm{H}.$$

pp2:

$$^{3}_{2}\mathrm{He} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{7}_{4}\mathrm{Be} + \gamma$$

$$^{7}_{4}\mathrm{Be} + e^{-} \rightarrow {}^{7}_{3}\mathrm{Li} + \nu_e$$

$$^{7}_{3}\mathrm{Li} + {}^{1}_{1}\mathrm{H} \rightarrow 2\ {}^{4}_{2}\mathrm{He}.$$

Here, the helium-4 produced in *pp1* starts direct interaction with helium-3. 31% of time *pp2* chain occurs.

pp3:

$$^{7}_{4}\mathrm{Be} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{8}_{5}\mathrm{B} + \gamma$$
$$^{8}_{5}\mathrm{B} \rightarrow {}^{8}_{4}\mathrm{Be} + e^{+} + \nu_e$$
$$^{8}_{4}\mathrm{Be} \rightarrow 2\,{}^{4}_{2}\mathrm{He}.$$

Capture of a proton by beryllium-7 nucleus. A proton is captured only 0.3% of the time in the center of the Sun. So probability of this reaction is very small.

The energy generation rate is determined the first reaction (intermediate deuterium production) since this is the slowest.

CNO Cycle:

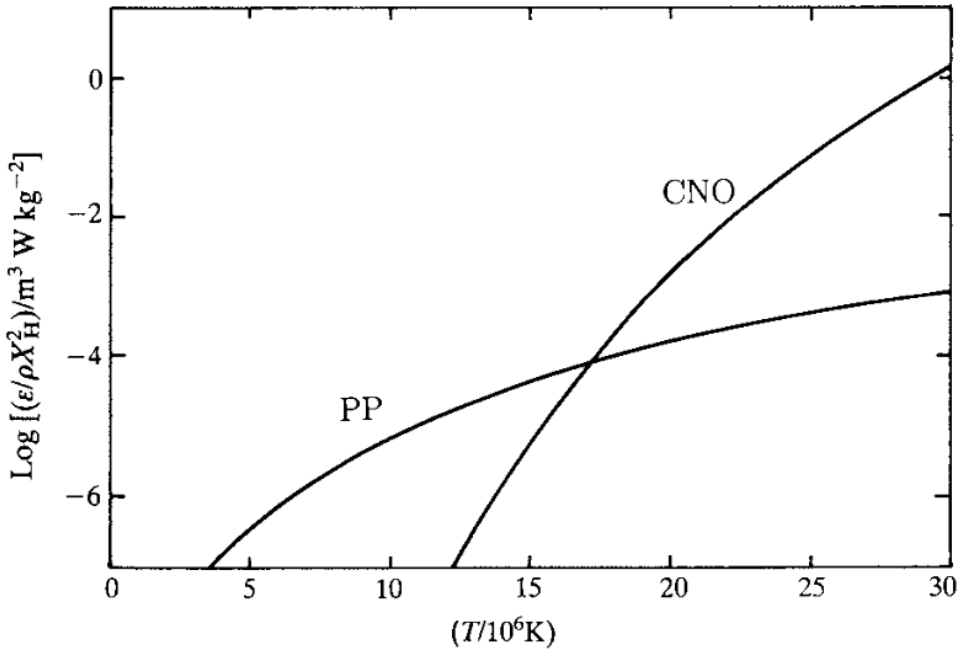
Independently suggested by von Weizsäcker (1938) and Bethe (1939).

If carbon, nitrogen and oxygen are present initially, they can act as catalysts (being consumed and then regenerated during the process). This reaction may also have two branches. The dominant one is shown below.

$$^{12}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{13}_{7}\mathrm{N} + \gamma$$
$$^{13}_{7}\mathrm{N} \rightarrow {}^{13}_{6}\mathrm{C} + e^{+} + \nu_e$$
$$^{13}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{14}_{7}\mathrm{N} + \gamma$$
$$^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{15}_{8}\mathrm{O} + \gamma$$
$$^{15}_{8}\mathrm{O} \rightarrow {}^{15}_{7}\mathrm{N} + e^{+} + \nu_e$$
$$^{15}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He}.$$

The reaction rate in the steady state is governed by the slowest reaction in the cycle, which in this case happens to be the fourth reaction

Comparison of energy generation rate in the *pp* chain and *CNO* cycle.



At temperatures below 20 million degrees the pp chain is the main energy production mechanism. At higher temperatures (corresponding to stars with masses above 1.5 $M_{\odot}$), the carbon (CNO) cycle becomes dominant, because its reaction rate increases more rapidly with temperature.

Helium to Carbon

The reaction sequence by which helium is converted into carbon is known as the **triple alpha** process.

$$^{4}_{2}\mathrm{He} + {}^{4}_{2}\mathrm{He} \rightleftharpoons {}^{8}_{4}\mathrm{Be}$$

$$^{8}_{4}\mathrm{Be} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{12}_{6}\mathrm{C} + \gamma.$$

Here beryllium 8 is unstable and decays into two helium nuclei or alpha particles in 2.6×10^{-16} seconds. The production of carbon thus requires the almost simultaneous collision of three particles.

Since this reaction involves three particles, it is much less likely to occur compared to reactions involving two particles. Also, the Coulomb repulsion is stronger between helium nuclei than between the nuclei involved in *pp* chain reactions, requiring a higher temperature.

Inside stellar cores, this reaction can take place when the temperature is higher than 10^{8} K.

Further reactions

In the high-temperature environment of helium burning, other competing processes are also at work. After sufficient carbon has been generated by the triple alpha process, it becomes possible for carbon nuclei to capture alpha particles, producing **oxygen**. Some of the oxygen in turn can capture alpha particles to produce **neon**. However, these reactions are fairly rare because of the ever higher Coulomb barrier (remember, we are talking of helium burning temperature, ~10^8 K) and thus are not important as stellar energy sources.

$$^{12}_{6}\mathrm{C} + ^{4}_{2}\mathrm{He} \rightarrow ^{16}_{8}\mathrm{O} + \gamma$$

$$^{16}_{8}\mathrm{O} + ^{4}_{2}\mathrm{He} \rightarrow ^{20}_{10}\mathrm{Ne} + \gamma$$

If a star is sufficiently massive, still higher central temperatures can be obtained and many other nuclear products become possible. Examples of available reactions include carbon burning reactions near 6×10^8 K,

$$^{12}_{6}\mathrm{C} + ^{12}_{6}\mathrm{C} \rightarrow \begin{cases} ^{16}_{8}\mathrm{O} + 2\,^{4}_{2}\mathrm{He} & *** \\ ^{20}_{10}\mathrm{Ne} + ^{4}_{2}\mathrm{He} & \\ ^{23}_{11}\mathrm{Na} + p^{+} & \\ ^{23}_{12}\mathrm{Mg} + n & *** \\ ^{24}_{12}\mathrm{Mg} + \gamma & \end{cases}$$

*** are endothermic (i.e., energy is absorbed, rather being released.)

Oxygen burning reactions near 10^9 K.

$$
{}^{16}_{8}\mathrm{O} + {}^{16}_{8}\mathrm{O} \rightarrow \begin{cases} {}^{24}_{12}\mathrm{Mg} + 2\,{}^{4}_{2}\mathrm{He} \quad *** \\ {}^{28}_{14}\mathrm{Si} + {}^{4}_{2}\mathrm{He} \\ {}^{31}_{15}\mathrm{P} + p^{+} \\ {}^{31}_{16}\mathrm{S} + n \\ {}^{32}_{16}\mathrm{S} + \gamma \end{cases}
$$

After several intermediate steps the burning of silicon produces nickel and iron.

$$
\begin{aligned}
{}^{28}\mathrm{Si} + {}^{28}\mathrm{Si} &\rightarrow {}^{56}\mathrm{Ni} + \gamma\,, \\
{}^{56}\mathrm{Ni} &\rightarrow {}^{56}\mathrm{Fe} + 2\,\mathrm{e}^{+} + 2\,\nu_{\mathrm{e}}\,.
\end{aligned}
$$

Elements heavier than iron are almost exclusively produced by *neutron capture* during the final violent stages of stellar evolution.

Solar nutrino problem

Neutrino is a by-product in many of the reactions that are believed to be taking place inside the Sun.

Since neutrinos interact with matter only through the weak interaction, most of the neutrinos created at the centre of the Sun would come out without interacting with the material of the Sun at all. Thus, at the Earth, we expect a flux of neutrinos directly coming from the centre of the Sun.

Detecting this flux of neutrinos is a sure way of confirming that nuclear reactions are indeed taking place in the centre of the Sun.

In the 1960s the famous first solar neutrino experiment began (Davis, Harmer and Hoffman, 1968). The flux of neutrinos was detected, but the experimentally measured flux was found to be about one-third of what was theoretically predicted.

The Davis neutrino detector contained 615,000 kg of cleaning fluid, C2Cl4 (tetrachlorethylene) in a volume of 377,000 liters (100,000 gallons). One isotope of chlorine is capable of interacting with neutrinos of sufficient energy to produce a radioactive isotope of argon that has a half-life of 35 days.

$$^{37}_{17}\mathrm{Cl} + \nu_e \rightleftharpoons {}^{37}_{18}\mathrm{Ar} + e^-.$$

The threshold energy for this reaction is 0.814 MeV and only the reaction in the pp3 involving ^{8}B can produce neutrino with this energy.

John Bahcall (a colleague of Davis) predicted that the experiment should have yielded a capture rate of 7.9 SNU (**solar neutrino unit)** while the actual data gave an average of 2.56 ± 0.16 SNU

Many other follow-up experiments were conducted: Japan's underground Kamiokande and Super-Kamiokande observatory (used water and able to detect ^{8}B neutrinos).

The Soviet–American Gallium Experiment (SAGE), located at the Baksan Neutrino Laboratory (inside a mountain in the Caucasus), and GALLEX (at the Gran Sasso underground laboratory in Italy) measure the low-energy pp chain neutrinos that dominate the Sun's neutrino flux.

SAGE and GALLEX make their detections via a reaction that converts gallium into germanium,

$$\nu_e + {}^{71}_{31}\mathrm{Ga} \rightarrow {}^{71}_{32}\mathrm{Ge} + e^{-}.$$

All these experiments confirm the deficit of neutrinos first established by the Davis detector. People started doubting the solar model.

Theoretical solution:

An elegant solution to the solar neutrino problem proposed that the solar model is essentially correct but that the neutrinos produced in the Sun's core actually change before they reach Earth. The **Mikheyev–Smirnov–Wolfenstein** (or **MSW**) **effect** involves the transformation of neutrinos from one type to another. This idea is an extension of the *electroweak theory* of particle physics that combines the electromagnetic theory with the theory of weak interactions governing some types of radioactive decay.

The neutrinos produced in the var- ious branches of the pp chain are all electron neutrinos (ν_e); however, two other *flavors* of neutrinos also exist—the muon neutrino (ν_μ) and the tau neutrino (ν_τ).

The MSW effect suggests that neutrinos oscillate among flavors, being electron neutrinos, muon neutrinos, and/or tau neutrinos during their passage through the Sun.

The neutrino oscillations are caused by interactions with electrons as the neutrinos travel toward the surface. Because the chlorine (Davis), water (Kamiokande and Super-Kamiokande), and gallium detectors (SAGE and GALLEX) have different threshold energies and they are sensitive only to the electron neutrino, their results were determined to be consistent with the MSW theory.

Such neutrino oscillations have been confirmed recently by the Sudbury Neutrino Observatory (Ahmad et al., 2002).

The nuclear reactions in the Sun produce electron neutrinos and all the solar neutrino experiments also detect electron neutrinos only. Presumably, during the flight from the Sun to the Earth, some of the electron neutrinos get converted into the other types and are not detected in the solar neutrino experiments. This is now believed to be the reason why the measured flux is less than what is theoretically predicted.